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Experiment, Modeling, and Simulation of Advanced Materials - Plasma
Interactions in the Space Environment

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14. ABSTRACT This project primarily aims to advance the understanding of the physics of plasmas composed of multi-species molecular gases, with a specific focus on investigating and developing techniques for creating Very Low Earth Orbit (VLEO)-like conditions in the laboratory to support airbreathing propulsion schemes and propulsion system testing. Key accomplishments include developing an analytical model for ExB probes to quantify the impact of probe design on measured ion velocity distribution functions (IVDFs), accounting for how the finite pass-band filter alters the true IVDF, and identifying that longer collimator and drift tube lengths are crucial for resolving closely adjacent peaks. A dual-collector ExB probe was designed and built at UIUC to enable simultaneous measurements with different drift tube lengths, revealing that while a short drift tube can merge adjacent peaks, a too-long tube might filter out smaller ones, and initial analytical IVDF reconstruction from this data was unsuccessful, leading to a shift towards numerical approaches for obtaining the transmittancy matrix. Simultaneously, the project modeled and simulated neutral beam generation via charge-exchange (CEX) reactions as a method to replicate VLEO conditions, with preliminary OD analysis suggesting approximately 30% neutralization of an argon ion beam is feasible. High-fidelity CHAOS simulations are underway to optimize gas cell designs for CEX, confirming the code's accuracy for unneutralized ion beam expansion, though current CEX ion densities from simulations (approximately $7.56 \times 10^{15} \text{ m}^{-3}$) are considered too low to match VLEO neutral number densities, largely due to CEX ions moving radially or backward because of slow axial velocity and Coulomb forces.					
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I. Accomplishments

The goal of this project is to further our understanding of the physics of plasmas consisting of multi-species molecular gases. In particular, the focus of this project is on investigating and exploring processes and techniques for creating VLEO-like conditions in the laboratory. This type of capability and the associated physics can be beneficial for investigating airbreathing propulsion schemes and for simulating and creating VLEO relevant conditions in the laboratory for testing and investigating propulsion systems within this environment.

Details of accomplishments during this reporting period:

1) major activities;

- Explored effects of ExB probe design and geometry on measured ion velocity distribution
- Compared measured IVDF vs. synthetically defined true IVDF
- Explained the ion collection in the ExB probe based on the ion beam trajectory within the given ExB probe geometry
- Developed the analytical model that connects the ion beam's true ion velocity distribution function (IVDF) and the measured ExB probe spectrum
- Applied the analytical model and explored the effects of ExB probe design and geometry on the deviation of the measured ExB probe spectrum from the true IVDF
- Simulated the RF plasma source in a grounded computational environment by introducing argon neutrals and ions and compared the results with experimental data.
- Modeled the charge-exchange process responsible for generating CEX ions, which was used to predict the population of fast neutrals.
- Performed the preliminary 0D analysis on the feasibility of the neutral beam generation from the ion beam by enhancing the charge-exchange reaction based on the known rate coefficients

2) specific objectives;

- Create multi-species chemically reactive gas plasmas relevant to chemical propellant decomposition product mixtures
- Develop the plasma diagnostic tools that can evaluate the multi-species chemically reactive gas plasmas
- Create neutral jets of gases similar to ambient conditions for airbreathing EP
- Develop the neutral beam diagnostic tools that can evaluate the generated neutral jets
- Develop collision models of multi-ion species in the CHAOS/PIC code.
- Modify PIC buffer boundary condition of multipole plasma chamber to accommodate multiple ion species.
- Explore the sensitivity of sheath formation to different ion masses and longer pulse widths to study plasma sheath dynamics coupled to electron motion.
- Initiate PIC simulations for low-orbit-altitude airbreathing or ambient atmosphere propulsion such as N_2^+ , N^+ , O^+ , O_2^+ in the PIC module of CHAOS.

3) significant results or key outcomes, including major findings, developments, or conclusions (both positive and negative);

- Developed an analytical model for quantifying the effect of ExB probe parameters on the measured ExB probe spectrum
- Developed the dual-collector ExB probe that allows two ExB probe measurements with two different ExB probe geometries (i.e., two different performances) in the single test
- Explored design approaches to enable multi-species ion beams to be discerned and identified from measured probe spectrum

4) other achievements. Include a discussion of stated goals not met.

- Recruited new PhD student Cody Myers to join the project
- Recruited new MS student Ron Hus to join the project
- Results were presented and disseminated at the 2024 AIAA SciTech conference
- Results were presented and disseminated at the 38th International Electric Propulsion Conference, Toulouse, France
- Results were presented and disseminated at the 2025 AIAA SciTech conference

II. Impacts

Development of the principal discipline(s) of the project

Future DoD Space Force missions must extend from very low Earth orbit (VLEO) to low Earth and geosynchronous Earth orbits (LEO/GEO), and to the cis lunar environment. These missions would benefit from spacecraft that are adaptable, flexible, and reconfigurable to meet a wide range of mission capabilities and requirements. Further, these missions would benefit from spacecraft that can gather and use in-situ resources, like Earth atmosphere gases. Propulsion concepts and approaches that have been shown to have promise to enable and enhance missions in these domains are multimode propulsion, airbreathing electric propulsion, and electrosprays. For example, some of our recent research has highlighted the flexibility and adaptability of multimode propulsion systems to meet a wide range of mission needs for robotic servicing scenarios that enable space logistics and mobility across multiple domains. An example of the mission capability space (or mission capability envelope) of multimode propulsion versus traditional stand-alone and hybrid (chemical and electric propulsion with separate propellant) propulsion approaches is shown in Figure 1.

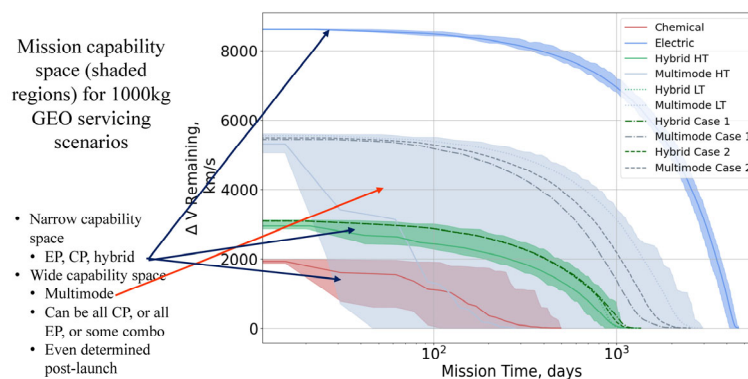


Figure 1: Example of the life history of the robotic servicing mission capability of a 1000 kg GEO servicer spacecraft with different types of propulsion. The multimode approach has a mission capability space (grey shaded region) that is 10x larger than stand-alone or hybrid propulsion.

Future development and successful implementation of advanced propulsion approaches, like multimode propulsion, must leverage molecular gas propellants. Specifically, future propulsion approaches must create and accelerate ions from plasmas formed using molecular gases and mixtures of molecular gases, such as air (N_2/O_2), N_2 , N_2/H_2 mixtures, and $\text{CO}_2/\text{H}_2\text{O}/\text{N}_2$ mixtures representative of the decomposition products of chemical monopropellants. **The problem** is that the physics of molecular gas plasmas like these is not well-understood, and is therefore not easily or readily implemented in advanced space propulsion concepts.

The **goal** of this project is to further our understanding of the physics of plasmas consisting of multi-species molecular gases. In particular, the focus of this project is on exploring the physics relevant to creating VLEO and airbreathing EP relevant flight-like conditions for future investigations and studies of such systems under relevant conditions and environment.

Describe the impact in this reporting period on the development of human resources

The project has had impact on development of human resources. It has provided opportunities for research in the field of plasma physics, plasma diagnostics, and plasma sources and ion beam sources. Specifically it has enabled the PIs and graduate students to further their knowledge and understanding of plasma physics and these other topics.

Describe the impact in this reporting period on physical, institutional, and information resources that form infrastructure.

The project has had impact on physical, institutional resources that form infrastructure. Specifically it has supported the development of experimental capabilities for creating plasma and interrogating plasmas of relevance to AFRL and society at large.

III. Changes

Changes in approach

None to report.

Problems or delays

None

Expenditure Impacts

None

Significant changes in the use or care of human subjects, vertebrate animals and/or biohazards

None

Changes to the primary place of performance from that originally proposed

None

IV. Technical Updates

In order to understand the plasma conditions in the experiment, we have developed the plasma diagnostic probes (a Faraday cup probe, a Langmuir probe, and an ExB probe).

An ExB probe is a pass-band velocity filter for an ion beam. When ions are accelerated by the same acceleration potential and gain the same kinetic energy, which is approximately true for many EP thrusters, the velocity of the ion can be estimated by $v_{ion} = \sqrt{2q_k V_{accel}/m_k}$. Therefore, unlike other diagnostic devices such as an RPA or an ESA, an ExB probe can discern ions with different mass-to-charge ratios, and an ExB probe is often used to determine the relative fractions of the multiply-charged ions in the noble gas plasma. Further improvements in the velocity resolution is critical for multi-species, chemically reactive plasma measurements because the ions in such plasma usually have similar mass-to-charge ratios. An ExB probe typically sweeps the filter velocity to obtain the ion velocity distribution function (IVDF). In fact, several ion parameters, such as the peak and average ion velocity, the ion energy spread, the structure of the ion acceleration region, and the relative ion fraction, are inferred from the measured IVDF. However, due to the finite pass-band filter window size, it has been known that the shape of the measured IV curve (i.e., the ExB probe spectrum) is not the same as the shape of the true IVDF [1,2].

Figure 2 [3] shows the operating mechanism of the ExB probe as a velocity filter. E- and B-fields are placed in the filter region to generate electrostatic and Lorentz forces on the ions. When these two forces are equal and cancel each other, the ions pass through the filter region. When one force is stronger than another, the ion drifts in either the positive or negative y-direction and is filtered out by the applied ExB fields. However, when the velocity is just slightly above or below the characteristic velocity, the ion drifts but can still pass through the filter region. This means that the ExB probe overcounts these ions, and that causes the difference in the measured ExB probe spectrum and the true IVDF.

We developed the ion collection model in the ExB probe by accounting for these ions. Figure 3 [3] shows the ion beam in the filter region deflected by Δx_f and Δy_f . The effective area of the ion beam is now the intersection of two circles (i.e., the apertures at the ends of the filter region) and the area ratio of the ion beam is called a transmittancy (T_k). Since the deflections of the ions at each section are determined by the velocity of ion (v_{ion}) and the characteristic velocity of the ExB probe (v_w), the transmittancy can be represented as a 2D matrix (T_k).

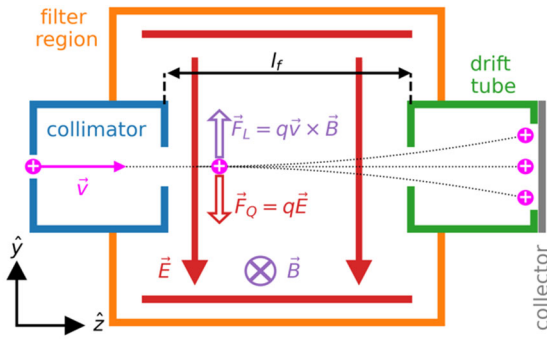


Figure 2: ExB probe Operating Mechanism [3]

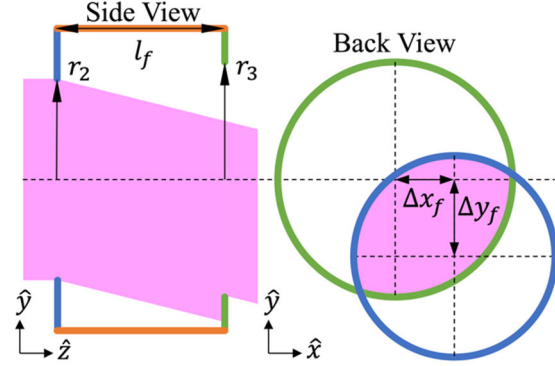


Figure 3: Concept of Transmittancy [3]

The ion current collected by the ExB probe at one electrode potential can be expressed as

$$I_{c,k}|_{V=V_{plates}} = A_1 q_k n_k \int_{v_w - \Delta v_w}^{v_w + \Delta v_w} \left[T_k|_{v_w = \frac{V_{plates}}{B_x d_e}} v f_k \right] dv \quad (1) [3]$$

Since the Transmittancy should be zero when the ion velocity is outside of the allowed velocity range ($v_w \pm \Delta v_w$), the integral in Eq. (2) [3] can be expressed in the form of the matrix multiplication.

$$I_{c,k}|_{V=V_{plates}} = A_1 q_k n_k T_k|_{v_w = \frac{V_{plates}}{B_x d_e}} v_{ion} f_k \quad (2) [3]$$

The collected ion current can also be represented as

$$I_{c,k}|_{V=V_{plates}} = A_1 q_k n_k v_w g_k|_{v_w = \frac{V_{plates}}{B_x d_e}} \quad (3) [3]$$

where g_k is the shape of the measured ExB probe spectrum. Comparing Eqs. (2) [3] and (3) [3],

$$g_k = \frac{v_{ion}}{v_w} T_k f_k \quad (4) [3]$$

Eq. (4) [3] explains how the true IVDF is altered to the measured ExB probe for the given ExB probe geometry, because T_k is a function of the ExB probe's design including the E- and B-fields.

By using the above relation, three ExB probe designs are tested for the one, synthetically-defined IVDF and how each design affects the ExB probe spectrum. The test IVDF is a mixture of three ions (Ar^+ , N_2^+ , and N^+) and each ion's IVDF follows the Gaussian distribution. We consider that the ions are accelerated by the same acceleration potential of 500 V, and the peak ion velocity is calculated from it. The peak width is 5% of the peak velocity. The peak height is adjusted so that the ion's number density ratio is 1:1:1. Three ExB probe designs are summarized in Table 1 [3]. Figure 4 [3] shows the angular-averaged transmittancy matrix for each ion for the ExB probe Design 1, and Figure 5 [3] shows the angular-averaged transmittancy matrix for Ar^+ ion for each ExB probe design.

Table 1: Tested ExB probe design parameters [3]

Parameter	Design 1	Design 2	Design 3	Unit
r_1	4.0	4.0	4.0	mm
r_2	4.0	4.0	4.0	mm
r_3	4.0	4.0	4.0	mm
r_4	4.0	4.0	4.0	mm
l_c	135.0	135.0	65.0	mm
l_f	152.4	152.4	152.4	mm
l_d	135.0	135.0	65.0	mm
B_x	0.129	0.144	0.129	Tesla

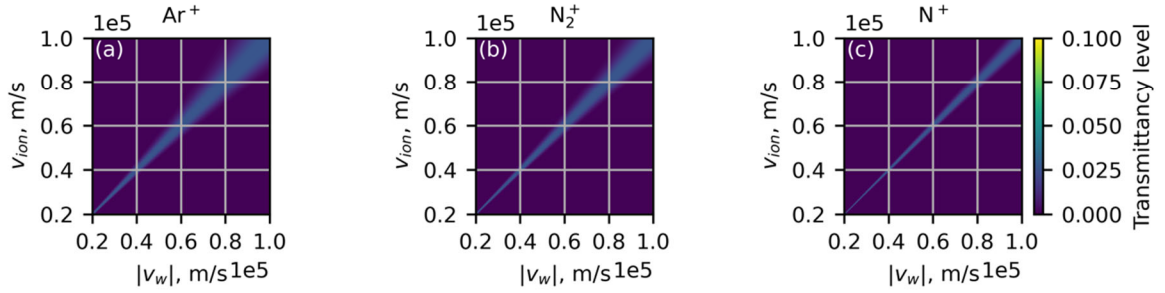
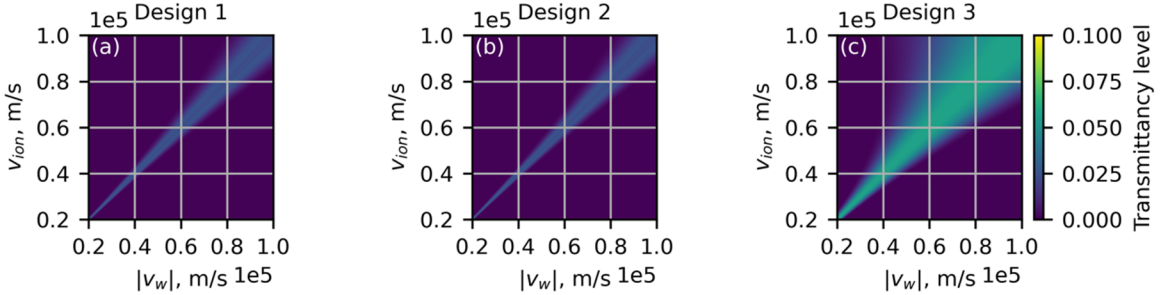
**Figure 4: Transmittancy for different ions for Design 1 ExB probe [3]****Figure 5: Transmittancy for different Designs for Ar+ [3]**

Figure 6 [3] shows the test IVDF (black line) and the calculated ExB probe spectrum for each design (colored lines). The shapes of the ExB probe spectra from Design 1 and Design 2 are similar because they share the same geometric parameters. However, since Design 2 overestimates the B-field strength, it underestimates the ion velocity. Unlike those two, Design 3 fails to capture the true IVDF. The ExB probe Design 3 broadens all three peaks and makes two distinct, closely adjacent peaks (Ar^+ and N_2^+) into one large peak. The only difference between Design 1 and Design 3 is the length of the collimator and the drift tube, and it indicates that these sections have to be long enough to resolve two adjacent peaks.

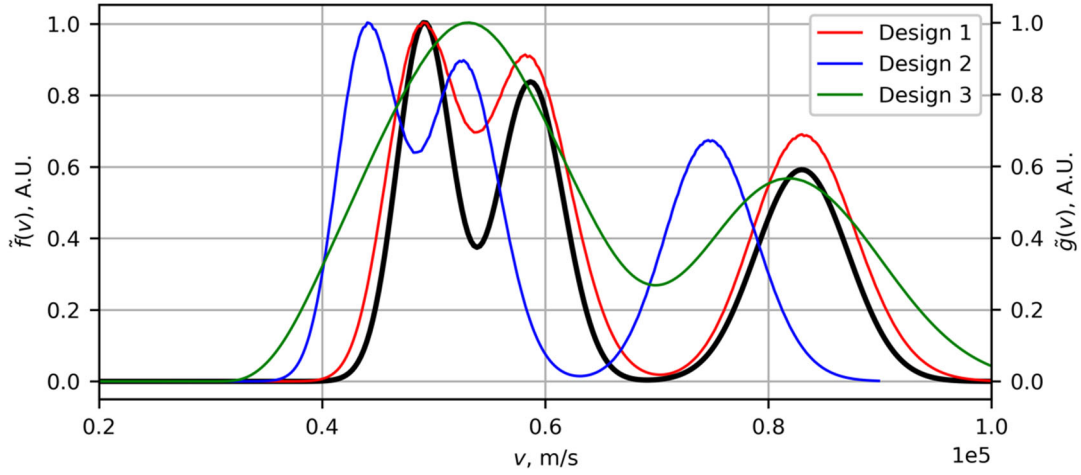


Figure 6: Tested IVDF and the calculated ExB probe spectra for different ExB probe designs [3]

The ion deflection in the filter region and the drift tube section is determined by the E- and B-fields. Although previous ExB probe analysis assumes uniform E- and B-fields, the actual E- and B-fields in typical ExB probes are not uniform, as shown in Fig. 7 (a) [3].

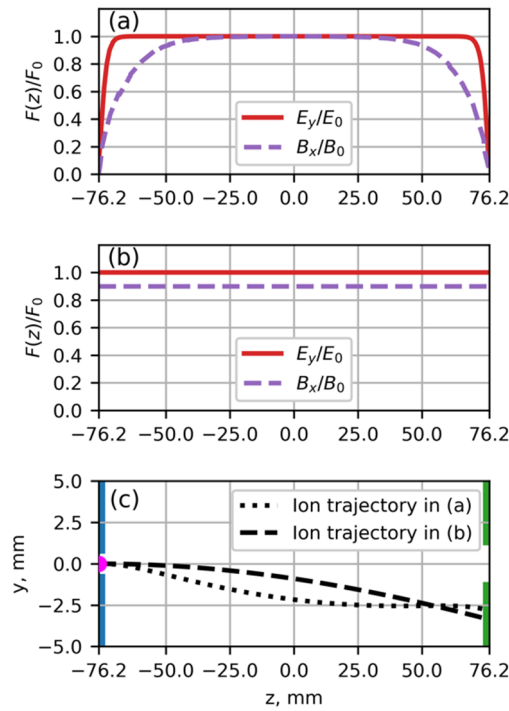


Figure 7: Evaluating the field non-uniformity effect. (a) Non-uniform E- and B-field, (b) Uniform E- and B-field, (c) the ion trajectories under the non-uniform and the uniform fields [3]

We proposed to use the practical B-field strength that is given by

$$B_{x,pra} = B_{x,0} \frac{B_{x,eff}/B_{x,0}}{E_{y,eff}/E_{y,0}} = B_{x,eff} \frac{E_{y,0}}{E_{y,eff}} \quad (5) [3]$$

where $B_{x,eff}$ and $E_{y,eff}$ are the effective field strength that is given by the area under the curve and $B_{x,0}$ and $E_{y,0}$ are the field strength at the center of the domain. The use of the practical B-field is tested by comparing the ion deflection under the non-uniform field (Fig. 7 (a) [3]) and the uniform field with the practical B-field (Fig. 7 (b) [3]). The ion trajectory is computed by using the 2D single ion particle motion code and by analytically solving the equation of motion. Figure 8 [3] compares the computed deflections. The analytical formula using the $B_{x,pra}$ captures the ion deflections under the non-uniform field with an error of -1.0 mm (approximately 30% percentage difference) or less for the displacement and -750 m/s (approximately 33% percentage difference) or less for the velocity change compared to the simulation.

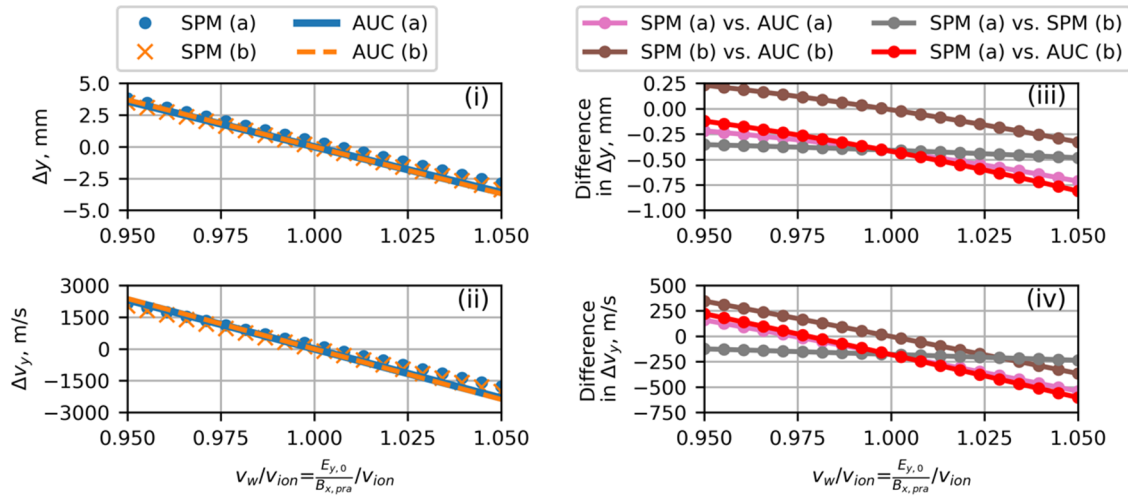


Figure 8: Evaluation the ion deflection calculation by the analytical model and by the single-ion trajectory [3]

Using the analytical approach discussed above, the ExB probe was designed, built, and tested at the UIUC EP Lab. Figure 9 shows the top view of the UIUC ExB probe (it is actually the dual-collector ExB probe discussed below, but the majority of parts are the same). The UIUC ExB probe is about 35 lbs and the central box that contains the two N42 permanent magnets and two electrodes sizes about 5.5 x 4.5 x 7.0 inch³. The 10-cm collimator has an inner diameter of 6 mm with no apertures at both ends (the collimator is essentially tube). The drift tube is also 10 cm and has an inner diameter of 6 mm. Figure 10 shows the collimator and the drift tube. The collector attached to one end of the drift tube is made of molybdenum and shaped in the cone-in-cylinder to minimize the secondary-electron emission effect on the collector. Figure 11 shows the collector parts before assembly. The collector is connected to the Keithley 6485 Picoammeter to measure the deposited current on the collector. The two electrodes, made of aluminum, are connected to two Matsusada R4G120 programmable DC power supplies. The maximum potential difference between two electrodes is 240 V (= +120 - (-120) V), which is equivalent to about 121 km/s for the given B-field strength (0.199 Tesla as the practical B-field strength) and the electrode spacing distance (10 mm). Figure 12 shows the Transmittancy matrix for the N₂⁺ and N⁺ ions and Figure 13 shows the collected IV curves by the UIUC ExB probe for the 200W HET operating with N₂.

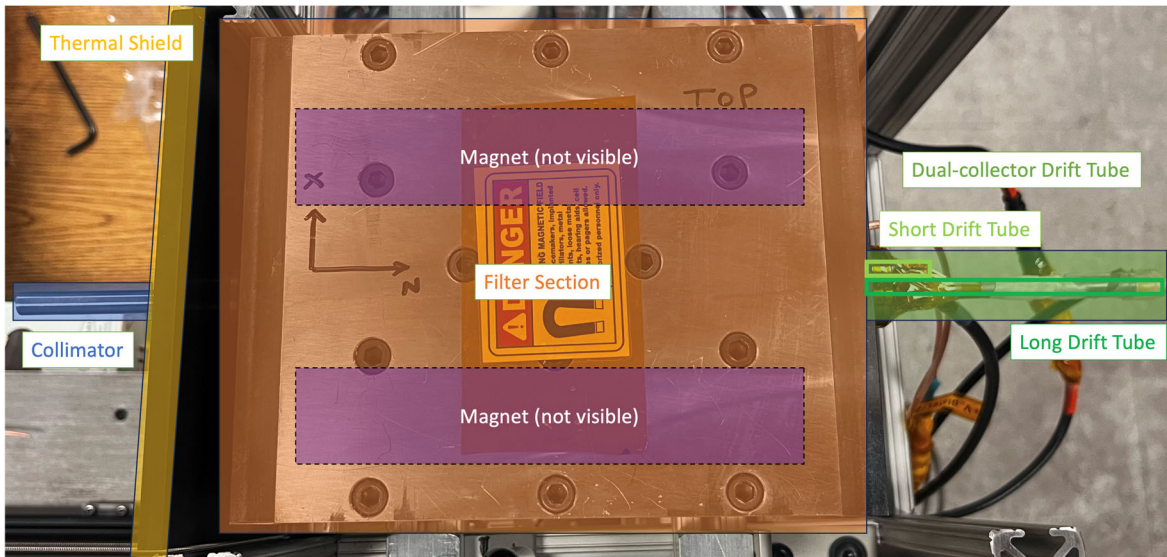


Figure 9: Top view of the dual-collector ExB probe developed at UIUC. Everything except the drift tube is identical to the UIUC ExB probe.



Figure 10: The collimator and the drift tube for the UIUC ExB probe.



Figure 11: The collector assembly in the UIUC ExB probe.

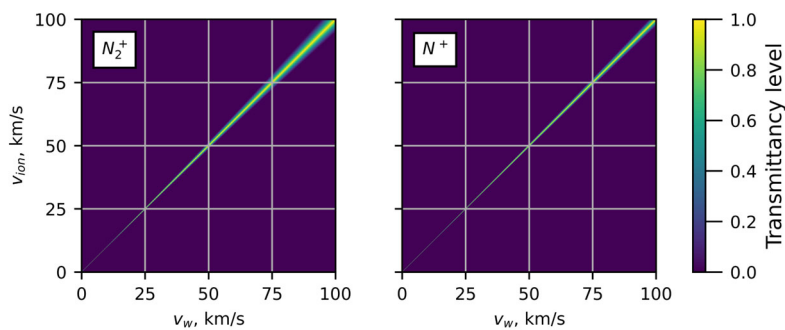


Figure 12L The transmittancy matrix for N_2^+ and N^+ for the UIUC ExB probe.

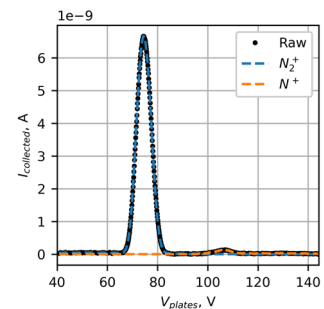


Figure 13: The raw measured data collected by the UIUC ExB probe.

In the traditional ExB probe, finding the optimal drift tube length may require multiple iterations. The developed analytical approach revealed how different ExB probe geometries affect the performance of the ExB probe. The length of the drift tube is the design parameter that can balance the pass-band width of the ExB probe and the amount of the collected ions (thus the SNR), and can be easily changed compared to the length of the central box, which is tied to the length of the permanent magnets. The dual-collector ExB probe developed at the UIUC has two semi-cylindrical drift tubes and allows us to make a measurement with two different drift tube lengths at the same time. **Figure 14** shows the dual-collector drift tube. The short drift tube length is 1.28 cm, while the long drift tube length is 10 cm. The difference in the drift tube length allows the ExB probe to have a completely different Transmittancy matrix, which allows us to collect different ExB IV curves for the same ion beam measurement. **Figure 15** shows the Transmittancy matrix for the long and short drift tubes for the N_2^+ ions.

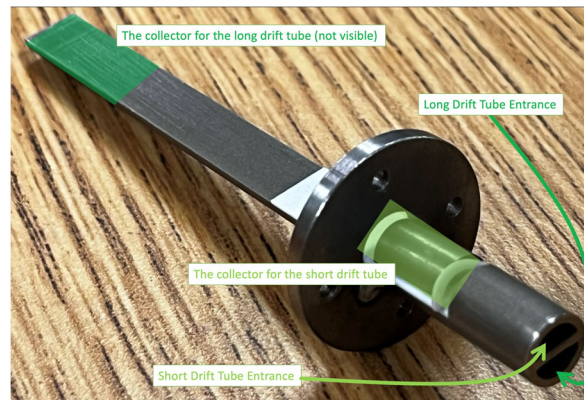


Figure 14: The design of the dual-collector drift tube.

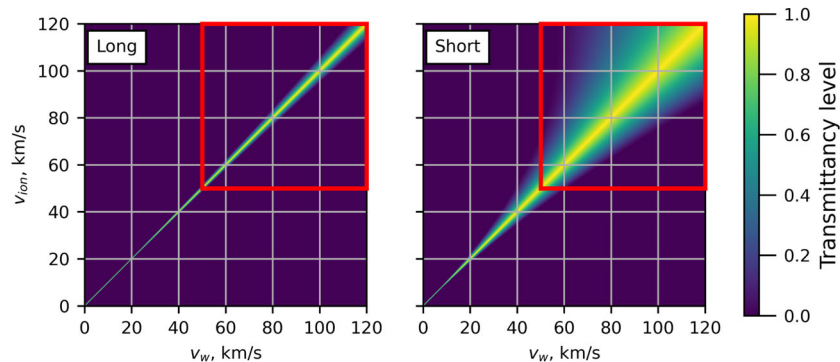


Figure 15: The Transmittancy matrix for the dual-collector ExB probe.

The dual-collector ExB probe was tested for the gridded ion thruster operating at a total acceleration potential of 800 V. After completing the data collection, we found a leak in the feed system, enabling us to determine the exact amount of flow used by the thruster. **Figure 16** shows the ExB IV curve collected by the dual-collector ExB probe when the ion thruster operated with one sccm molecular-nitrogen (and some amount of leaked air). As expected, more current was collected from the short drift tube ExB probe. However, the long drift tube ExB probe observed that there are more than two ion species, which was expected due to the large amount of air contributing to the total flow. The passband width for the short

ExB probe was large enough to make these two adjacent peaks into one large peak, and the smaller passband by the long drift tube revealed that there were more than two ion species in the ion beam. **Figure 17** shows the IV curve collected by the dual-collector ExB probe when the ion thruster operated with one sccm argon (and some amount of leaked air). In this case, the long drift tube ExB probe filtered out the ions enough to make it unable to observe the smaller peaks observed by the short drift tube ExB probe. It indicates that neither the short nor long drift tube lengths used in this setup were sufficient to properly study the ion beam, and it requires different approaches to obtain a good IV curve (e.g., by moving the ExB probe closer).

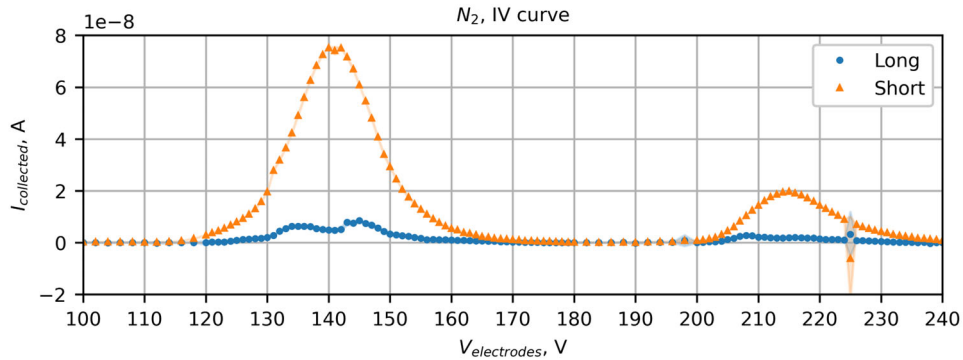


Figure 16: The raw data measured by the dual-collector ExB probe for N₂.

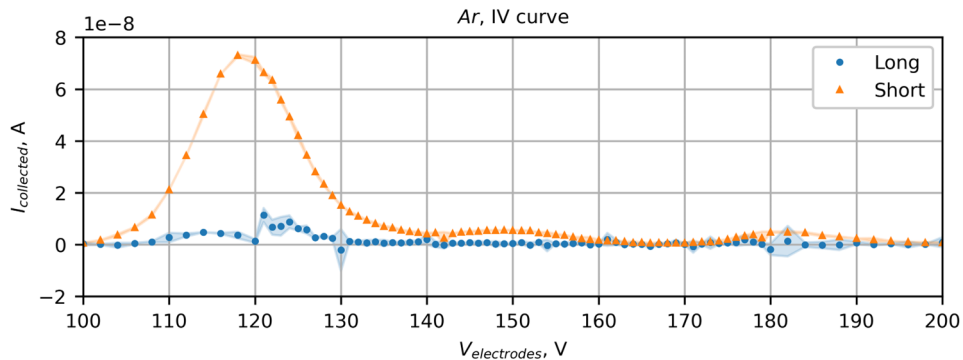


Figure 17: The raw data measured by the dual-collector ExB probe for Ar.

The initial efforts were made to reconstruct the IVDFs from the IV curves collected by the dual-collector ExB probe, with the analytically obtained Transmittancy matrix. It indicates analytical approach failed to accurately reflect the ion beam collection characteristics that occurred in the dual-collector ExB probe. The aperture size was too large compared to the electrode spacing distance, and the assumption of the uniform beam deflection within the aperture may not be accurate in this ExB probe. The current work focuses on obtaining the Transmittancy matrix from the numerical approach by tracking the single ion trajectory for many conditions.

An on-ground flight condition test has yet to be conducted for the air-breathing electric propulsion due to a lack of capability to generate an 8 km/s low-density neutral beam in a vacuum facility. Recently, several groups used the Hall effect thruster's quasi-neutral beam instead of a neutral beam to test the performance of the low-density molecular flow collection devices [4–7]. The current density from the typical HET is 1~10 mA/cm², which is $6.24 \times 10^{19} \sim 10^{20}$ #/m²s in flux (using $\Gamma = J_i/e$). This corresponds to an altitude of 189~139 km within the VLEO range. However, this approach ignores the possible acceleration or deceleration of the charged particles by the sheath formed on the air collection devices. Also, the operation of the HET at the low beam energy (~5 eV) is not standard in the EP community; lowering the anode voltage itself would not be enough because it then cannot give electrons energy enough for ionization. Studying the direct ionization of the flowing neutrals is impossible with this approach. At the University of Illinois Urbana-Champaign Electric Propulsion Laboratory, we aim to generate a neutral beam by neutralizing the ion beam by the charge-exchange (CEX) reactions. A gas cell is placed downstream of an ion beam source and an increased neutral number density in the gas cell enhances the CEX reaction. Fast ions entering the gas cell will maintain kinetic energy and leave as fast neutrals. This concept is similar to the neutral beam injection in the fusion field [8], and the neutral beam propulsion developed for an asteroid deflection [9–11]. In the neutral beam propulsion work, the 1 keV Ar⁺ ion beam was neutralized with 85% neutralization fraction. A beam of the atmospheric species (e.g., N₂⁺, O₂⁺, and O⁺) at a lower velocity (~8 km/s) needs to be neutralized to replicate the VLEO condition.

To verify the feasibility of a neutralizing gas cell leveraging resonant CEX reactions, as 0D model of the neutralization of an argon ion beam was created. Figure 18 shows the percent neutralization of an ion beam flux in a 1 mTorr gas cell from CEX reactions with neutral argon, where any ions that experienced course-altering MEX collisions were counted as losses. The model showed that approximately 30% of a 20 A/cm² ion beam could be neutralized in 10 cm without experiencing trajectory altering collisions.

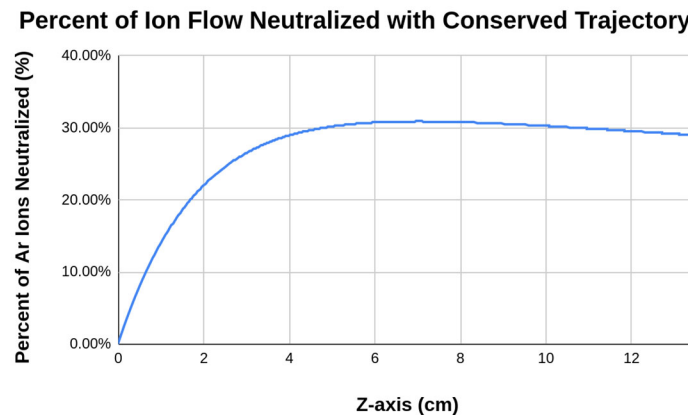


Figure 18: The percentage of 20 A/cm² argon ion beam neutralized in 1 mTorr gas cell over cell length.

Simulations for two competing designs of a neutralizing gas cell are being modeled using CHAOS, UIUC's PIC-DSMC simulator (Jambunathan and Levin, 2018). These high-fidelity simulations will be able to take into account the geometry of each design, electron behavior, and consider the 3D nature of the system, all of which the previous 0D simulation could not. Since the length and

pressure of the gas cell will affect how many CEX reactions occur, a high-fidelity simulation will also allow the design of the cell to be adjusted to optimize the expected performance without having to manufacture a new gas cell.

The two gas cell designs to be simulated are one using a thermal ion source and one using an ion source with a directed ion beam. Design one has an open 10cm radius inlet with a decelerating potential mesh to decrease the speed of an ion beam produced by a HET from 30km/s to 8km/s (a standard VLEO orbit speed). The quasi neutral mixture of ions and electrons flows through the 1mTorr, 20cm long pressurized gas cell, where CEX reactions will neutralize a portion of the flow. Leftover ions and electrons at the open outlet will be filtered from the flow with an applied electric field, leaving only the portion of the beam that was neutralized.

Design 2 seeks to reduce the number of species in the chamber, and therefore the number of undesired collisions, by filtering out any electrons in the flow with a large negative potential barrier at the inlet. This concept may work better with a dense thermal plasma that could be siphoned into the gas cell by this negative potential at the inlet. A second biased mesh after the initial electron barrier mesh would adjust the speed of the ions to the desired value of 8km/s. The ions beam would then enter the 1 mTorr gas cell, and rapidly expand into the cell due to the large potential gradient created by the unneutralized beam's own space-charge.

CHAOS has never been used to model a flow composed entirely of ions, so it was necessary to find an analytical or numerical equation to solve for an unneutralized ion beam envelope (the radius of the outermost ion trajectory from the center of an ion beam). Using the derivation of the Universal Beam Equation in "Microwave and RF Vacuum Electronic Power Sources" by Richard G. Carter, a numerical method of analysis was derived. The comparison of this derived numerical result to an ion beam vacuum expansion in CHAOS, as shown in Figure 19, showed that CHAOS models this system accurately.

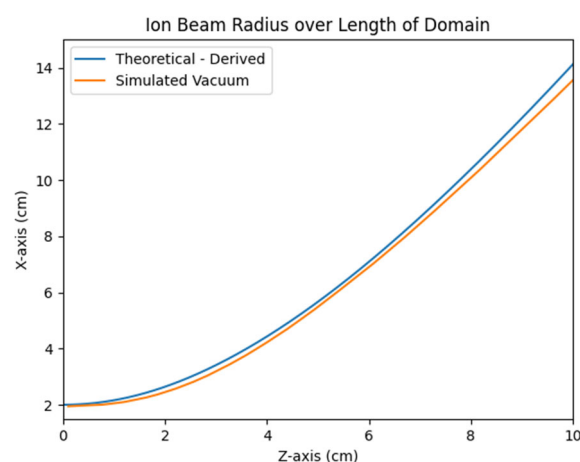


Figure 19: The trajectory of an ion along the outer edge of the unneutralized ion beam. The z-axis is the axis of beam propagation.

Our simulations were performed using the CHAOS code in house, the UDAA-based hybrid approach for octree simulation. It couples the Particle-In-Cell (PIC) method and the Direct Simulation

Monte Carlo (DSMC) method. PIC will be solving the Poisson equation. DSMC modeled collisions between ion-neutral and neutral-neutral species, thereby including the charge exchange process (CEX) and the momentum exchange process (MEX). The parallel computation technique is applied to save computational time. In the numerical setup, the calculation domain is modeled as a cubic region for simplicity. The ion current density is set at $0.01952 \text{ (A/m}^2\text{)}$, with an ion velocity of 8500 m/s . This information determines the ion number density using the relation to give a value of $1.4 \times 10^{13} \text{ m}^{-3}$. The neutral density is dependent on the background pressure, which ranges from 2.5×10^{-5} to $1.0 \times 10^{-4} \text{ torr}$, and the temperatures of both neutral and ion temperatures are assumed to be equal to the wall temperature of 325 K . Given this pressure range and temperature, the neutral number density varies from approximately $7.4 \times 10^{17} \sim 3.0 \times 10^{18} \text{ m}^{-3}$. The cell size Δx , must be smaller than λ_D , which for our conditions has a value $\lambda_D = 2.8 \times 10^{-3} \text{ m}$. The plasma frequency $\omega_{pe} = 2.11 \times 10^{-8} \text{ rad/s}$. Because the simulation did not introduce electrons, the timestep of the ions was allowed to be larger. Here, the ion timestep is $4.7 \times 10^{-8} \text{ s}$.

Table 2 provides detailed ion and neutral input parameters. When setting the timestep and the weighting factor, they have to meet the criteria, Eq. (6). The particle boundary conditions for all walls follow a Maxwellian distribution at 325 K . The electric boundary condition is set to Dirichlet with 0V . To decrease the calculation time and avoid not having enough memory for storing the neutrals, the calculation domain is cut in half at the center of the thruster since it is symmetric at the centerline. Therefore, the $x = 0$ wall is set as a specular boundary and Neumann boundary conditions. In addition, a numerical pump is set at the downstream corner. The size of the pump is $5 \times 5 \text{ cm}^2$. The reason for this setup is to avoid having an effect on the plasma sheath of the wall.

$$\nabla^2 \phi = -\rho/\epsilon_0 \quad (6)$$

$$E = -\nabla \phi \quad (7)$$

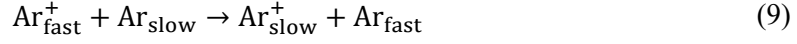
Table 2: Simulation species input setup.

Simulation Conditions	Ar	Ar ⁺
Current Flux (mA/m ²)	-	19.5
Number Density (m ⁻³)	7.4×10^{17}	1.4×10^{13}
Temperature (K)	325	325
Pressure (Torr)	2.5×10^{-5}	-
Bulk Velocity (m/s)	300	8500
Source Radius (cm)	-	0.25
Timestep (s)	4.7×10^{-5}	4.7×10^{-8}
Fnum	1×10^6	1×10^6
Weighting factor	0.5	0.0005
Input type	Half-Maxwellian	Half-Maxwellian

$$\frac{W_{ion} \Delta t_n}{W_{neu} \Delta t_i} = 1 \quad (8)$$

In order to create fast neutrals, the charge exchange (CEX) process is critical. The CEX process is an elastic collision between slow neutral and fast ions, shown in equation (9). The CEX cross-section is provided

in Eq. (10). Therefore, as 8500 m/s fast ion collides with a 300 m/s slow neutral, the velocity is switched, so the fast neutrals are created. The amount of CEX ions will create the same amount of fast neutrals as the generation happens. The results are shown below.



$$\sigma_{\text{CEX}} = 58.827 - 4.307 \ln(E_{ev}) \cdot 10^{-20} \text{ m}^2 \quad (10)$$

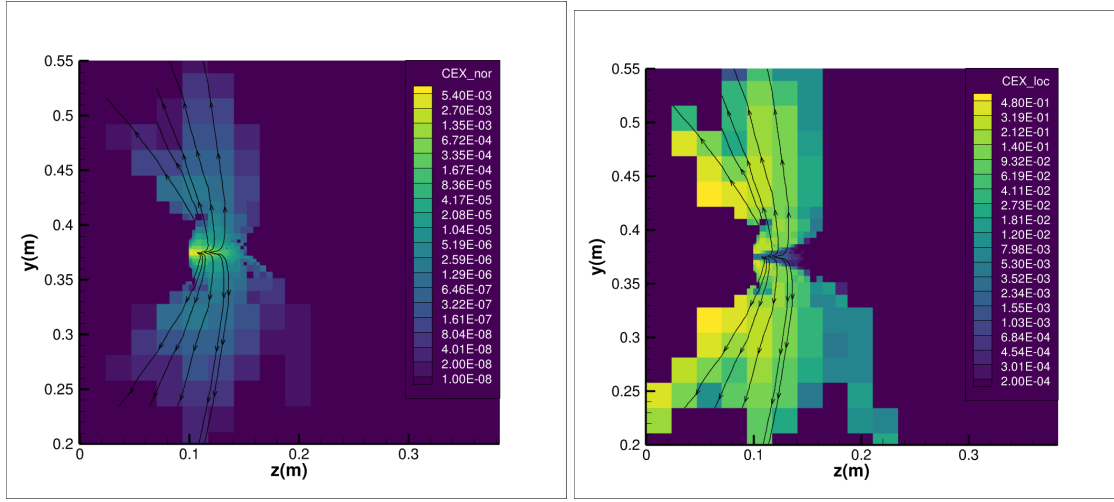


Figure 20: (a) The global normalized CEX ions (n_{CEX}/n_0) (b) The local normalized CEX ions ($n_{\text{CEX}}/(n_i + n_{\text{CEX}})$)

The charge exchange cross-section of the argon ions to create fast neutrals is given by Eq. (5). The results of the CEX ions can indicate that fast neutrals are created. Figure 20 shows two different types of normalized CEX ions. Fig. 20(a) is the CEX ions fraction normalized by the initial ion number density. The fraction at the thruster exit is about 0.54 % or multiplying this value with the initial ion density, $1.4 \times 10^{13} \text{ m}^{-3}$, the maximum CEX ion density is found to be $7.56 \times 10^{10} \text{ m}^{-3}$ at the thruster exit. Fig. 1(b) shows the CEX ions fraction of the local ions and CEX ions. In contrast to Fig. 20(a), most of the CEX ions move to the backflow region, but the fast ions just move directly to the downstream region. The fraction at the thruster exit is lowest, meaning that the fast ions are still the main species at the thruster exit. The CEX ions are slow, so they are easily affected by the Coulomb force, created by the electric field. This can be seen from the streamlines and both contour plots. The fraction of CEX ions can be up to almost 50 % at the backward region. When CEX ions are created, they immediately move radially due to the slow axial velocity. The CEX ions stop being created or moving to downstream after $z = 0.2 \text{ m}$ at the centerline.

To ensure that the fast ions have a sufficiently high velocity to create fast neutrals, the ion velocities for both radial and axial directions are also examined. Figure 21 shows both velocity components, demonstrating that the averaged fast ions maintain an axial bulk velocity of 8500 m/s as they exit the thruster, though the fast ions can be accelerated to 9200 m/s. Meanwhile, some ions exhibit backward and radial motion. This can be accounted for by the electric potential. The maximum potential, 2.3 V, is along the ion plume centerline. The potential difference creates electrical fields pulling the ions backward. The small potential difference along the ion plume centerline also means that the ions have no significant axial

acceleration, causing the maximum ion radial velocity to be around 8500 m/s having the opposite effect of charge exchange. Eq (5) shows that the cross-section is decreased followed by the increase of the kinetic energy calculated by the relative speed between targeted ions and neutrals. It also explains why most of the CEX ions are moving backward to the backflow region. As mentioned above, the amount of CEX ion density is equal to the amount of fast neutrals so we can expect $7.56 \times 10^{-10} \text{m}^{-3}$ are generated per second. However, this value would be too low compared with the number density of the VLEO environment.

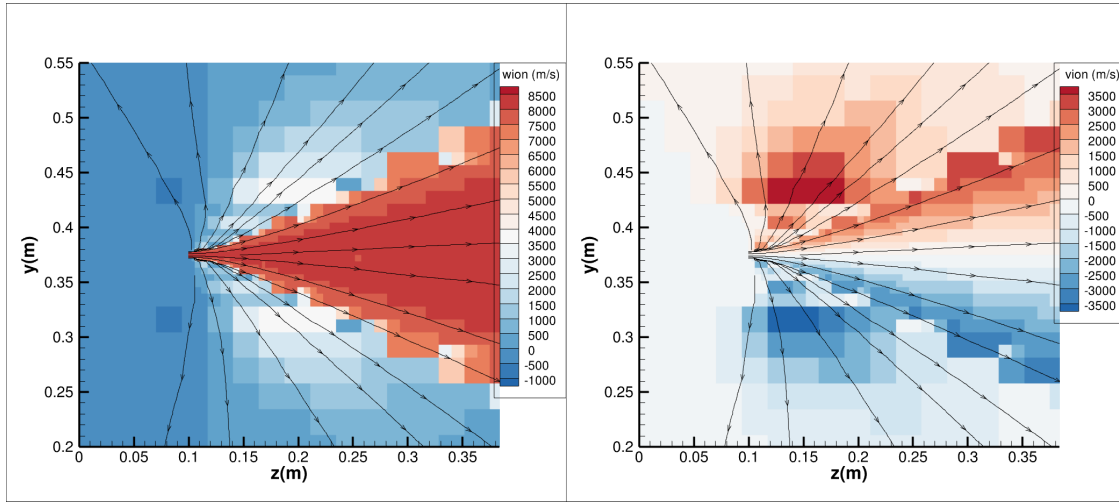


Figure 21: (a) The axial velocity component of ions. (b) The radial velocity component of ions.

Based on the experimental setup, argon ion data were obtained using ExB and Langmuir probes under varying background pressures, ranging from 2.5×10^{-5} to 8.0×10^{-5} Torr. Additionally, two oxygen cases were examined under background pressures of 1.0 and 5.0 sccm. As a result, Ar^+ , O^+ , and O_2^+ ions are considered as input species from the RF source. We will first focus on the argon cases.

In the argon case, both neutrals and ions are introduced at the RF source, with argon neutrals serving to maintain the background pressure. In subsequent simulations, electrons will be introduced to enhance ion beam collimation. This is expected to increase charge exchange (CEX) ion production and extend the neutral beam length. Simulations will also be conducted under different background pressures for comparative analysis. The simulation results will be evaluated against experimental data using ion velocity distribution functions (IVDFs) and peak ion energy measurements in the downstream region.